

Contents

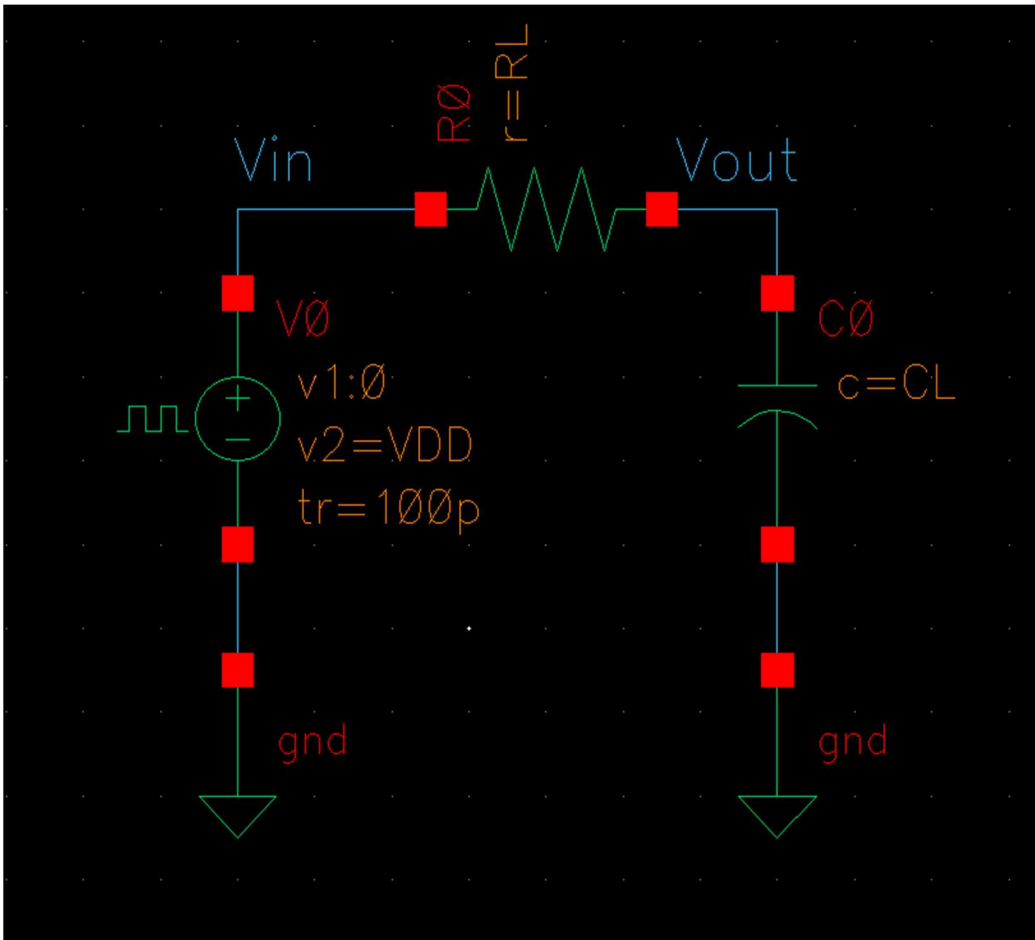
Part 1: Low Pass Filter	2
Transient Analysis	2
Schematic & Design.....	2
Input voltage properties	3
Time from 10% to 90% and vice versa	4
Analytical results comparison	4
Parametric Sweeps for R_l 1:1:5k	5
AC Analysis	6
Bode Plot.....	6
3db bandwidth and DC gain	7
Comparison	8
parametric sweep for $R=1,10,100,1000k\Omega$ Magnitude	8
Part 1: MOSFET Characteristics	10
ID vs VGS	10
Comment.....	11
gm vs VGS.....	12
Comment.....	12
ID vs VDS	13
Comment.....	13

Task 1

Part 1: Low Pass Filter

Transient Analysis

Schematic & Design



parameters
$C_L=0.5pf$
$R_L=1k\ ohm$
$T_{aw}=0.5ns$

Input voltage properties

CDF Parameter	Value	Display
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	0 V	off
Voltage 2	VDD V	off
Period	Tclk s	off
Delay time	Tclk/4 s	off
Rise time	100p s	off
Fall time	100p s	off
Pulse width	Tclk/2 s	off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off

OK Cancel Apply Defaults Previous Next Help

Result

50

Lower Limit 11

Upper Limit 100

Generate Clear

Square wave voltage

Tclk=50ns

Duty cycle 50%

Tfall=Trise=100ps

1. Transient Analysis Results

Transient Options

Time Step Algorithm State File Output EM/IR Output Misc

SIMULATION INTERVAL PARAMETERS

start

outputstart

TIME STEP PARAMETERS

step

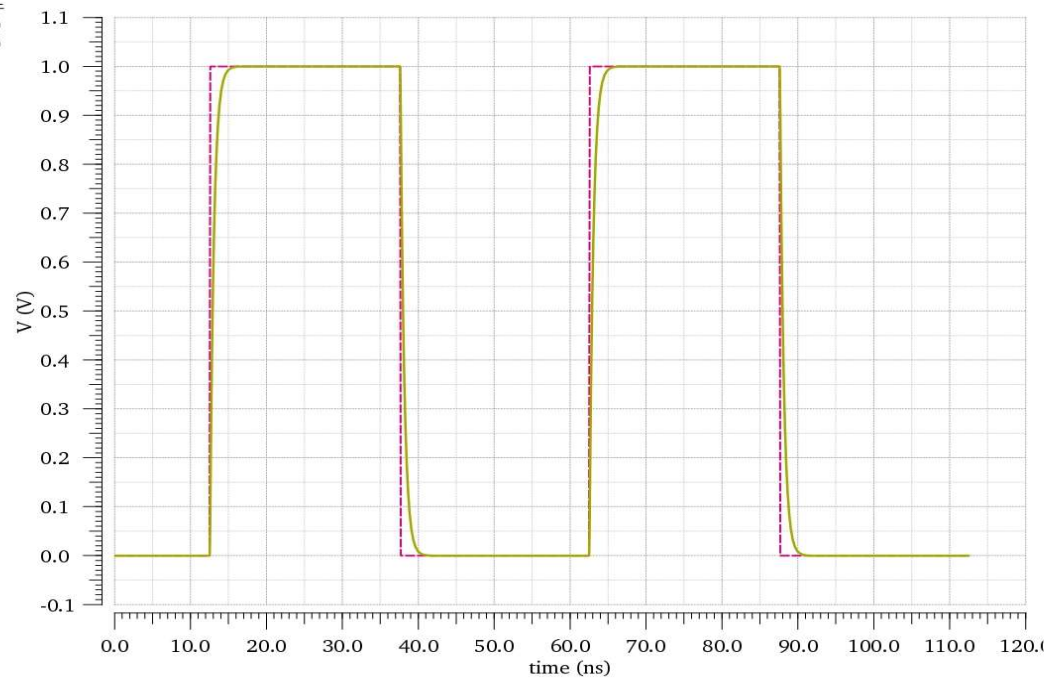
maxstep Tclk/100

minstep

OK Cancel Defaults Apply Help

Transient Analysis with a
max step of Tclk/100

Name	Vis	RL
/Vin		1000
/Vout		1000



Time from 10% to 90% and vice versa

lab1:lpf:1	riseTime(VT("/Vout") 0 nil 1 nil 1...	1.103 n
lab1:lpf:1	fallTime(VT("/Vout") 1 nil 0 nil 90...	1.103 n

Analytical results comparison

Using the first order response rule to calculate rise and fall time

$$y(t) = y_{\infty} + (y(0) - y_{\infty})e^{-t/\tau}, \tau = RC$$

So

$$\ln\left(\frac{y(t) - y_{\infty}}{y(0) - y_{\infty}}\right) = -\frac{t}{\tau}$$

Solving for t

$$t_{10 \rightarrow 90} = \tau \ln(9) \approx 2.2\tau$$

So t rise = 1.0986ns $\approx$ 1.1ns

$$t_{10 \rightarrow 90} = t_{90 \rightarrow 10}$$

And also t fall = 1.0986ns $\approx$ 1.1ns

Note : Here when we solved for t analytically we did not take into account the rise time and fall time of 100ps in calculation as we used the first order response equation and we assumed by this an ideal square wave

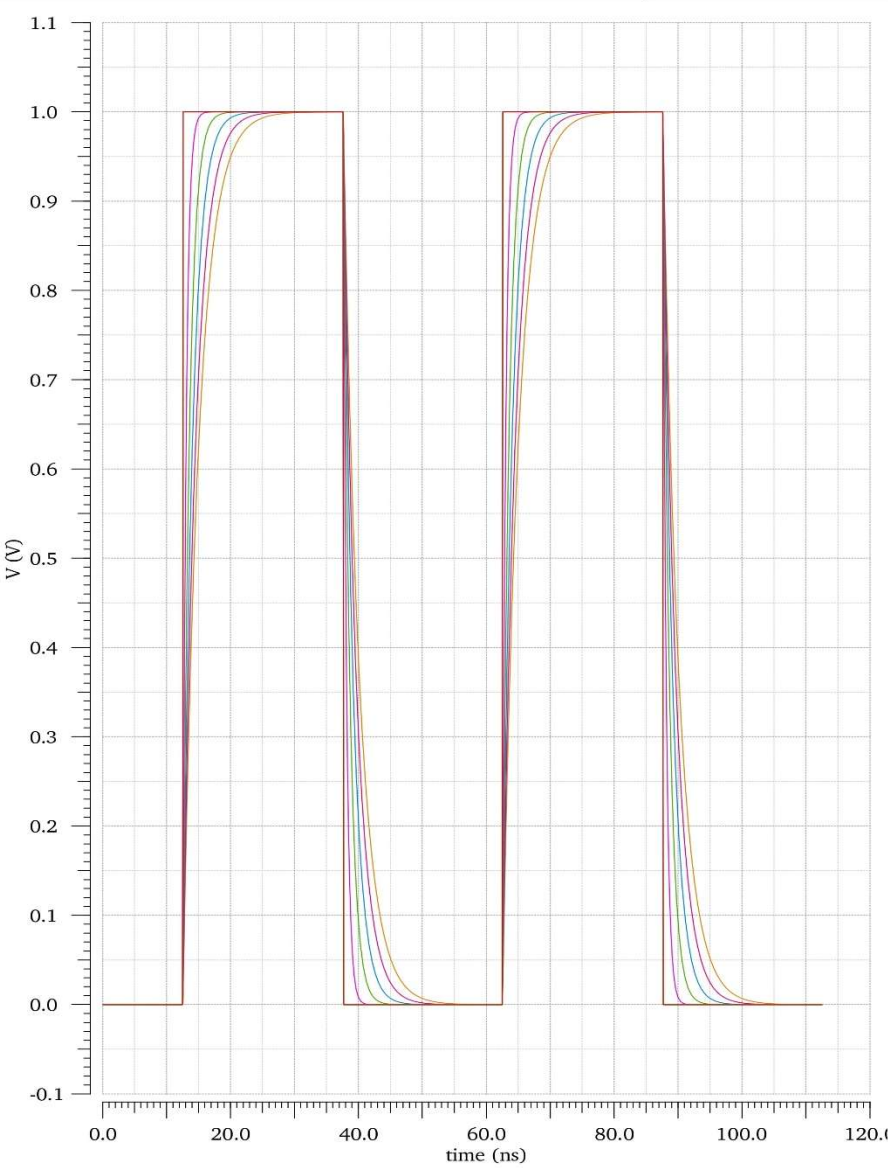
POC	Cadence	analytical
T Rise	1.103ns	1.1ns
T Fall	1.103ns	1.1ns

Parametric Sweeps for RL 1:1:5k

Transient Response

Sat Jun 20 12:41:30 2026

Name	Vis	RL
/Vin		
1000		
2000		
3000		
4000		
5000		
/Vout		
1000		
2000		
3000		
4000		
5000		



Point	Test	Output	Nominal
Parameters: RL=1 k			
1	lab1:lpf:1	riseTime(VT("/Vout") 0 nil 1 nil 1...	1.103 n
1	lab1:lpf:1	fallTime(VT("/Vout") 1 nil 0 nil 90...	1.103 n
Parameters: RL=2 k			
2	lab1:lpf:1	riseTime(VT("/Vout") 0 nil 1 nil 1...	2.197 n
2	lab1:lpf:1	fallTime(VT("/Vout") 1 nil 0 nil 90...	2.197 n
Parameters: RL=3 k			
3	lab1:lpf:1	riseTime(VT("/Vout") 0 nil 1 nil 1...	3.282 n
3	lab1:lpf:1	fallTime(VT("/Vout") 1 nil 0 nil 90...	3.282 n
Parameters: RL=4 k			
4	lab1:lpf:1	riseTime(VT("/Vout") 0 nil 1 nil 1...	4.383 n
4	lab1:lpf:1	fallTime(VT("/Vout") 1 nil 0 nil 90...	4.383 n
Parameters: RL=5 k			
5	lab1:lpf:1	riseTime(VT("/Vout") 0 nil 1 nil 1...	5.477 n
5	lab1:lpf:1	fallTime(VT("/Vout") 1 nil 0 nil 90...	5.477 n

Comment : the Rise and Fall time is proportional to the value of the Resistance as $\tau = RC$ and T fall or T Rise is proportional to τ

AC Analysis

Bode Plot

Magnitude

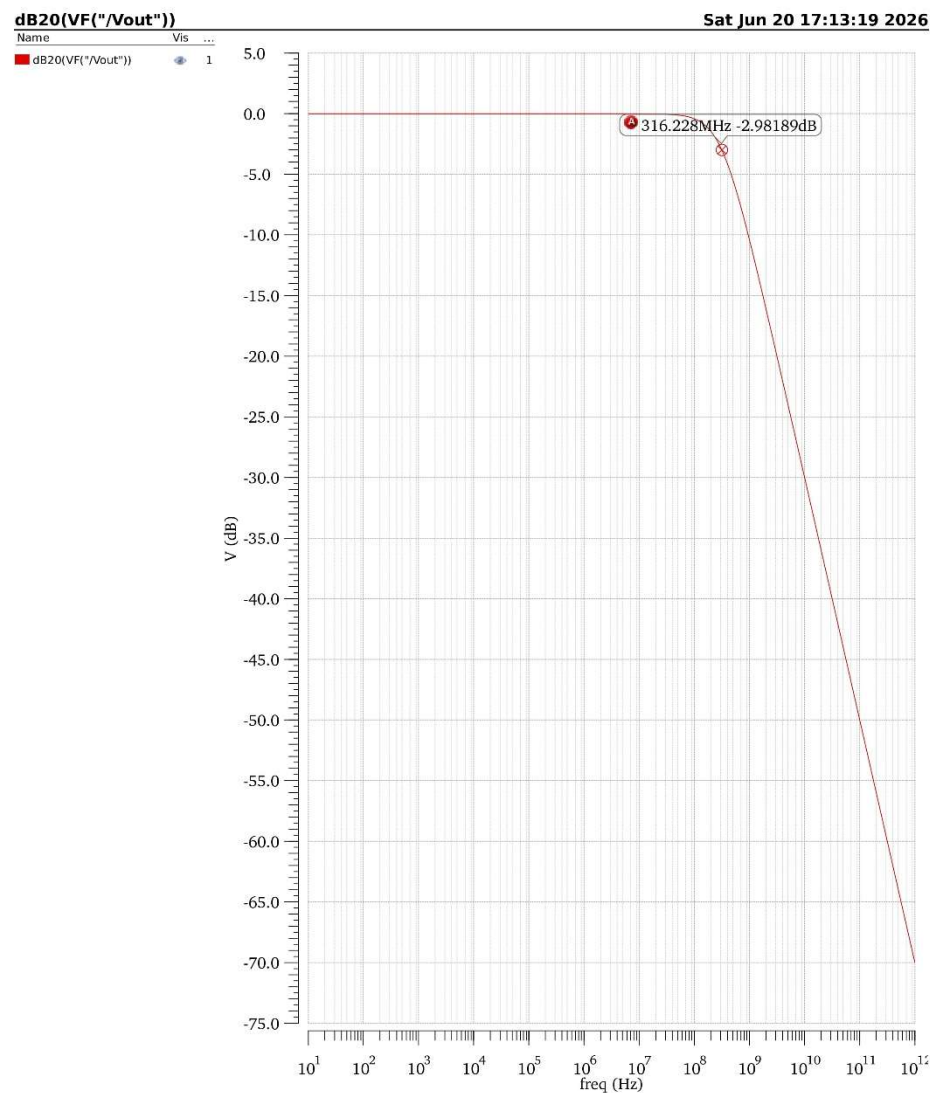
Cut off frequency

is 316MHz

$$= \frac{1}{2\pi RC}$$

at this frequency the

attenuation is -3db



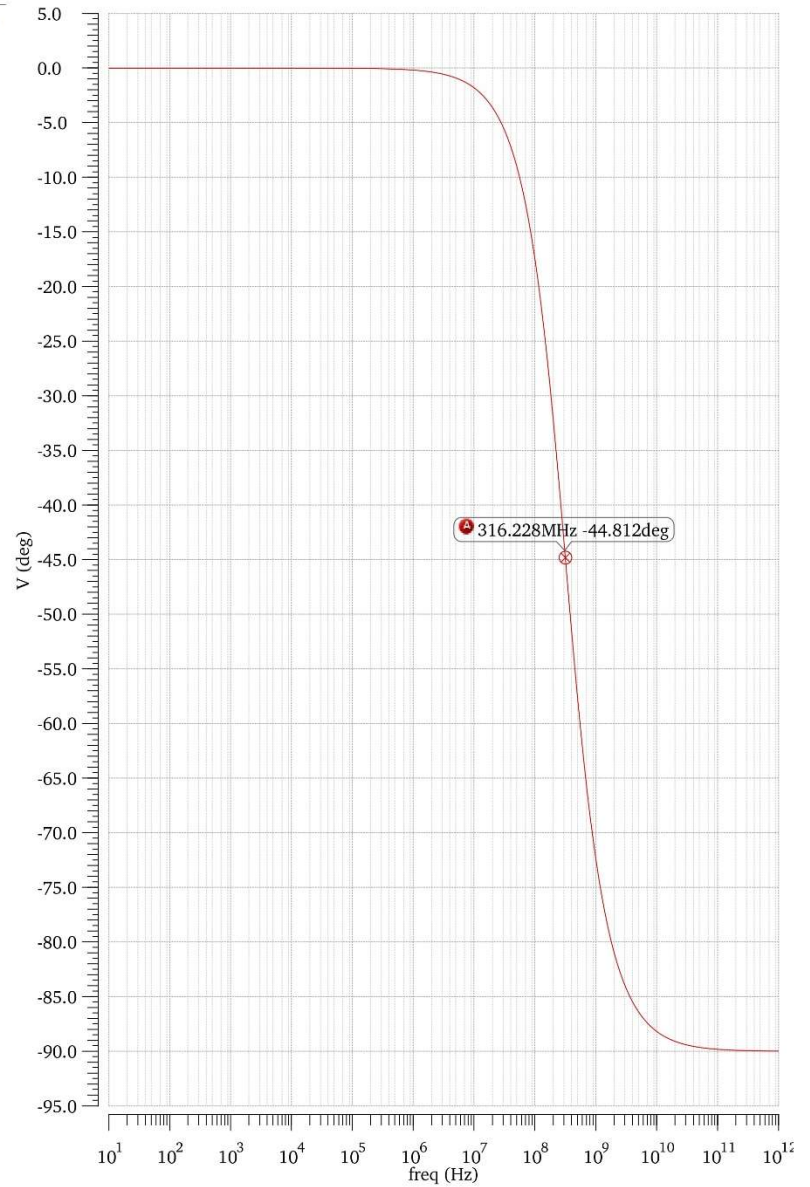
Phase

At the cutoff frequency the
Phase shift is -45 deg

phaseDeg(VF("/Vout"))

Sat Jun 20 17:13:19 2026

Name
phaseDeg(VF("/Vout"))



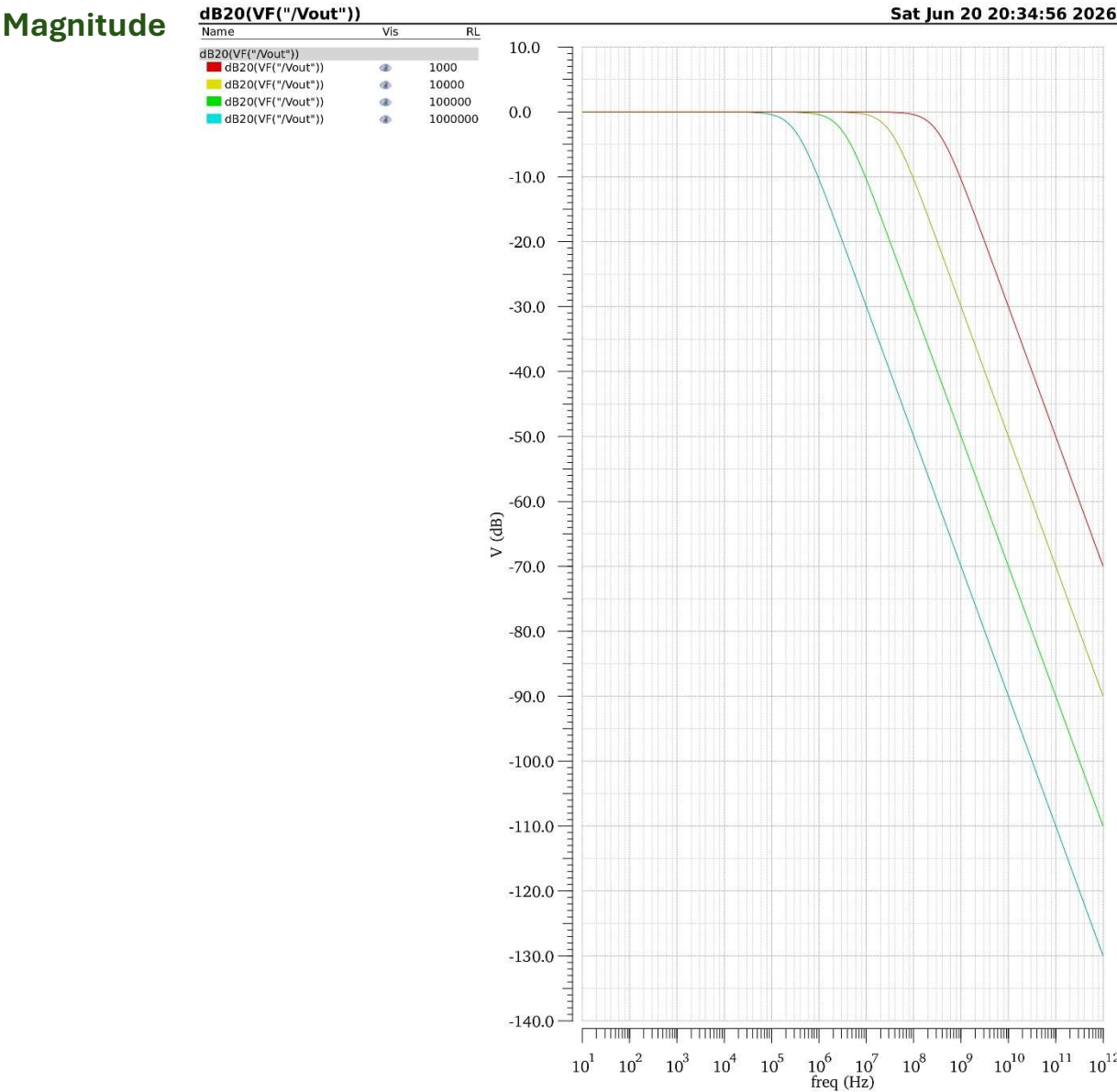
3db bandwidth and DC gain

lab1:lpf:1	BW	317.6M			
lab1:lpf:1	DC gain	1			

Comparison

POC	Simulation	Analytical
BW	317.6MHz	$= \frac{1}{2\pi RC} = 318.309\text{MHz}$
DC gain	1	$\frac{1}{\sqrt{(0.5 \text{ n} \cdot 0)^2 + 1}} = 1$

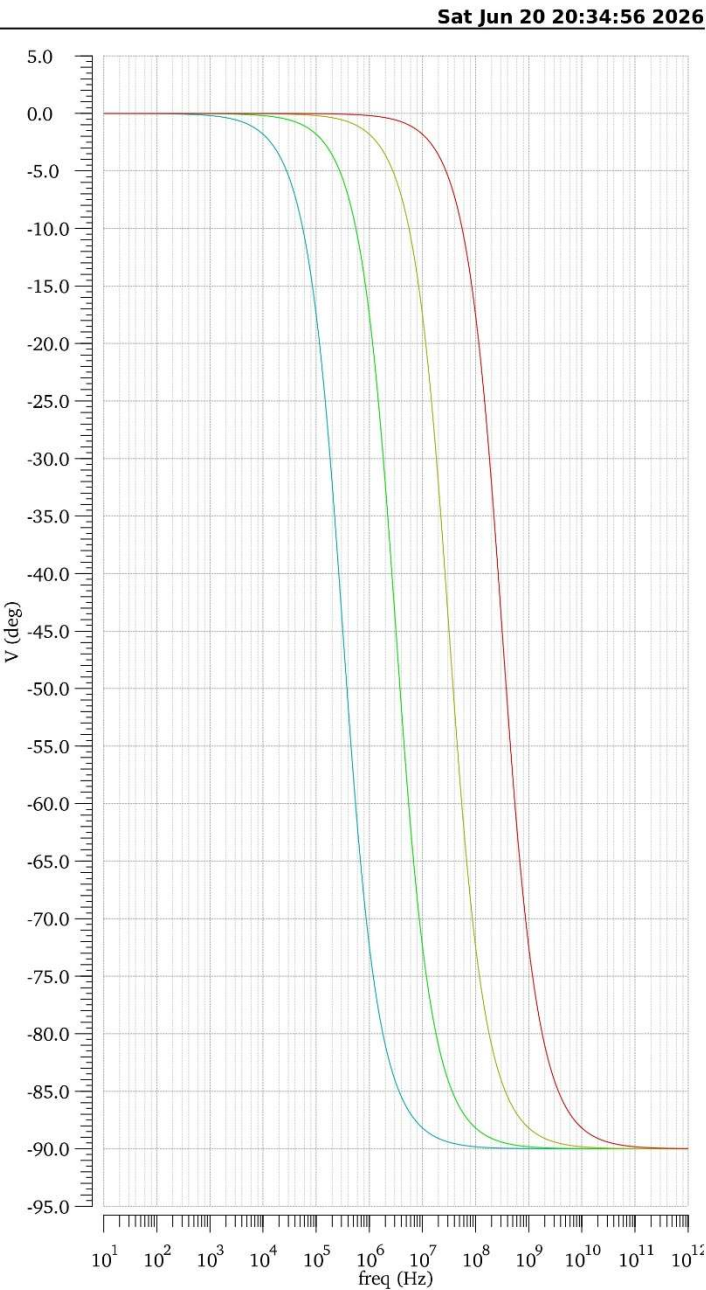
parametric sweep for $R=1,10,100,1000k\Omega$



Phase

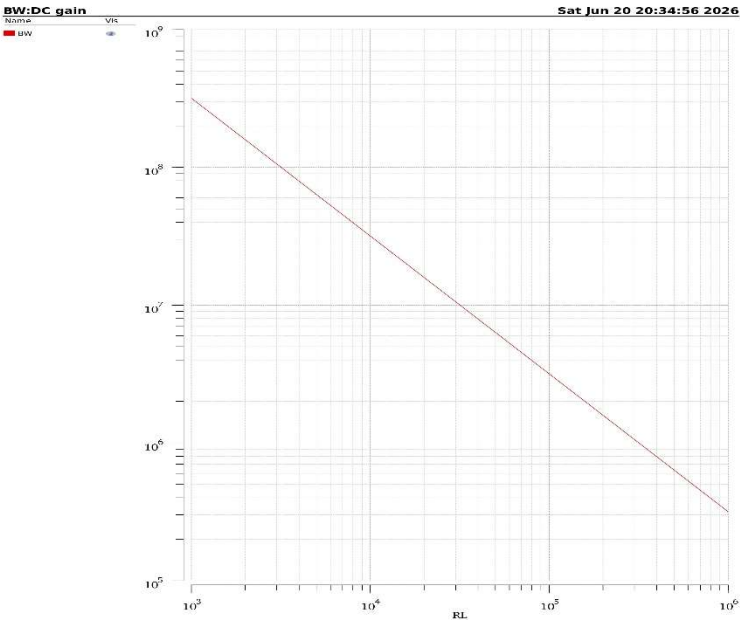
phaseDeg(VF("/Vout"))

Name	Vis	RL
phaseDeg(VF("/Vout"))		
phaseDeg(VF("/Vout"))		1000
phaseDeg(VF("/Vout"))		10000
phaseDeg(VF("/Vout"))		100000
phaseDeg(VF("/Vout"))		1000000

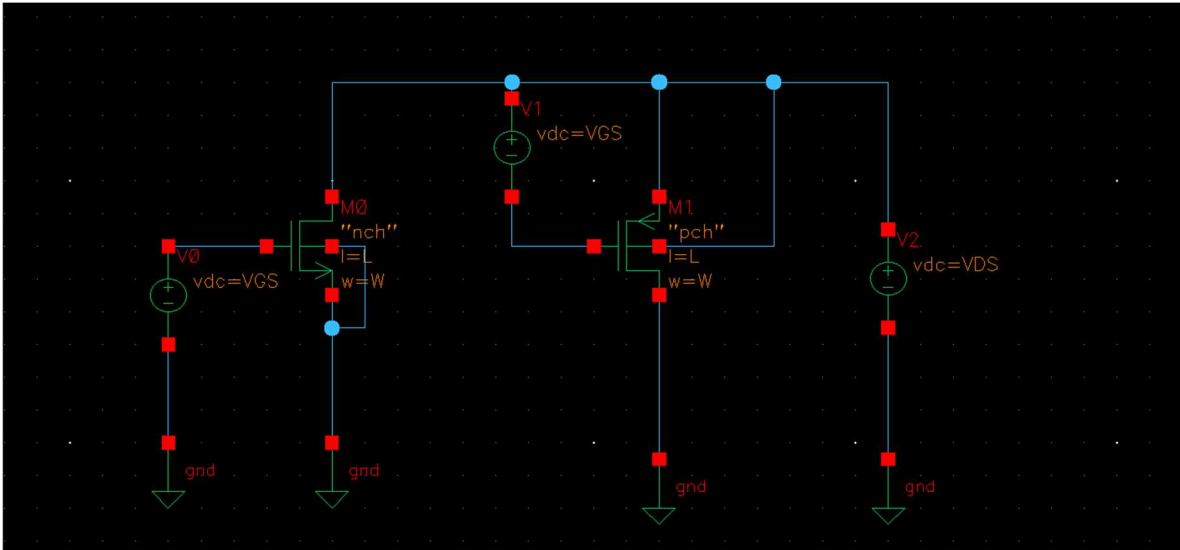


Point	Test	Output	Nominal
Parameters: RL=1 k			
1	lab1:lpf:1	BW	317.6M
1	lab1:lpf:1	DC gain	1
Parameters: RL=10k			
2	lab1:lpf:1	BW	31.76M
2	lab1:lpf:1	DC gain	1
Parameters: RL=100k			
3	lab1:lpf:1	BW	3.176M
3	lab1:lpf:1	DC gain	1
Parameters: RL=1M			
4	lab1:lpf:1	BW	317.6k
4	lab1:lpf:1	DC gain	1

Each time the resistance increases by a decade the BW or cutoff Frequency decrease by also a decade in other words when we multiply the RL by 10 the BW get divided by 10 and the phase roll off happens earlier



Part 1: MOSFET Characteristics



Channel	L	W
Short channel	200nm	3.6um
Long channel	2um	36um

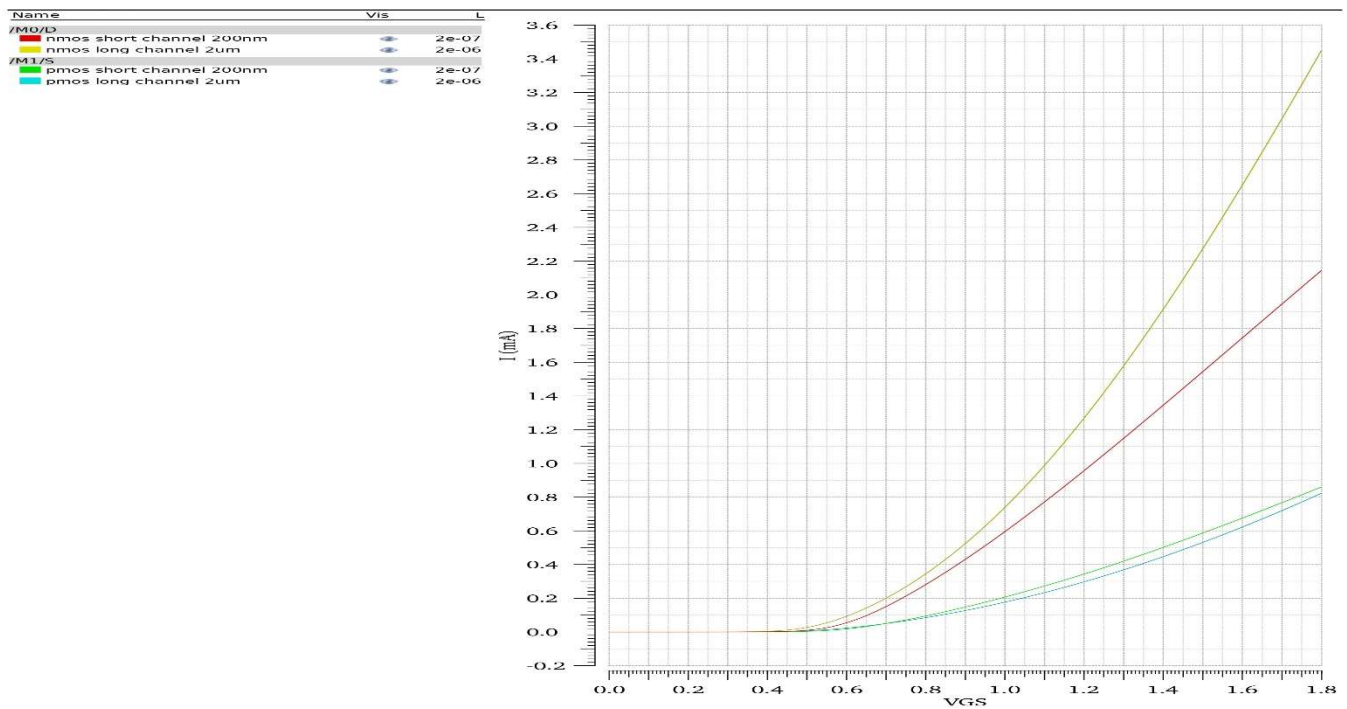
Result

18

Lower Limit

Upper Limit

ID vs VGS



Comment

Difference between short channel & long channel

1. As we can see it is clear that the long channel mosfet has higher current in comparison with the short channel one as we can see that at the beginning at low VGS they are close but at higher VGS the difference appear larger this is due to Velocity saturation and mobility degradation which are short channel effects which lead to less current at in short channel mosfet the electrons may hit velocity saturation before Pinch off which results in less overall current as we can see here
2. Here we can see that the relation at first is quadratic for both short channel and long channel but as the VGS increases the relation starts to be more linear for the short channel mosfet due to velocity saturation once Electric field critical is hit

Difference between Nmos & Pmos

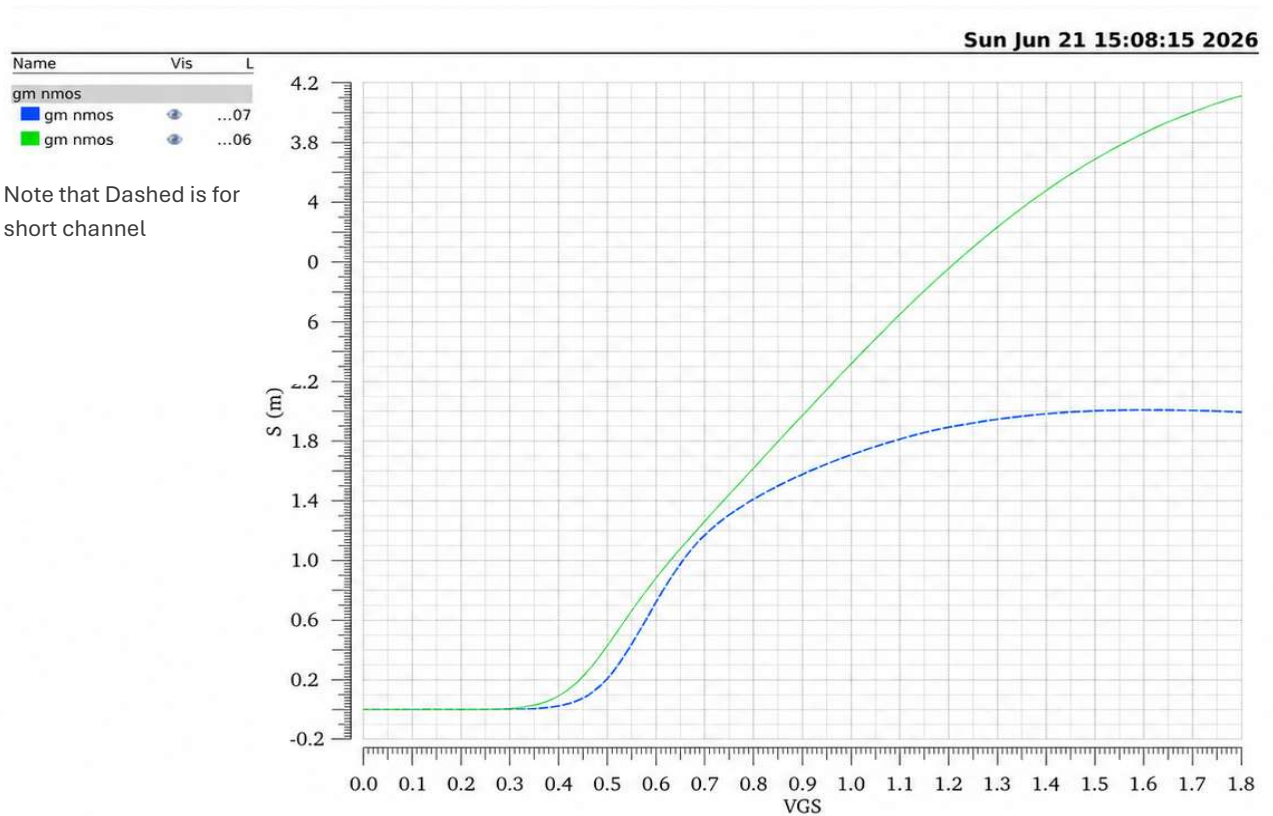
1. Nmos has higher current due to the electrons higher mobility than holes
2. Ratio between the current in Nmos and Pmos should be the the ratio between their mobilities as $I_D = \frac{1}{2} \mu_{n,p} C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$ so ID is directly with the mobility ($\mu_{n,p}$)

1	lab1:mosfet:1	L	200n			
1	lab1:mosfet:1	W	3.6u			
1	lab1:mosfet:1	ID max nmos	2.145m			
1	lab1:mosfet:1	ID max pmos	860.1u			
2	lab1:mosfet:1	L	2u			
2	lab1:mosfet:1	W	36u			
2	lab1:mosfet:1	ID max nmos	3.45m			
2	lab1:mosfet:1	ID max pmos	823.2u			

So as we can see here the Ratio between the currents at VGS=VDD is 4.2 for long channel and 2.5 for short channel

3. The Nmos is more affected by the short channel effects.

gm vs VGS



Comment

Linearity of gm

In long channel and in short channel before velocity saturation gm increase linearly as $g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$ so it is obviously linear in long channel and it is linear before it hit V_{sat} in short channel but after hitting V_{sat} in short channel gm saturates and becomes constant

Saturation of gm

In short channel yes it saturates as the relation become $g_m = K V_{DSsat}$ due to velocity saturation but in long channel it doesn't hit V_{sat} so the gm don't saturate in long channel.

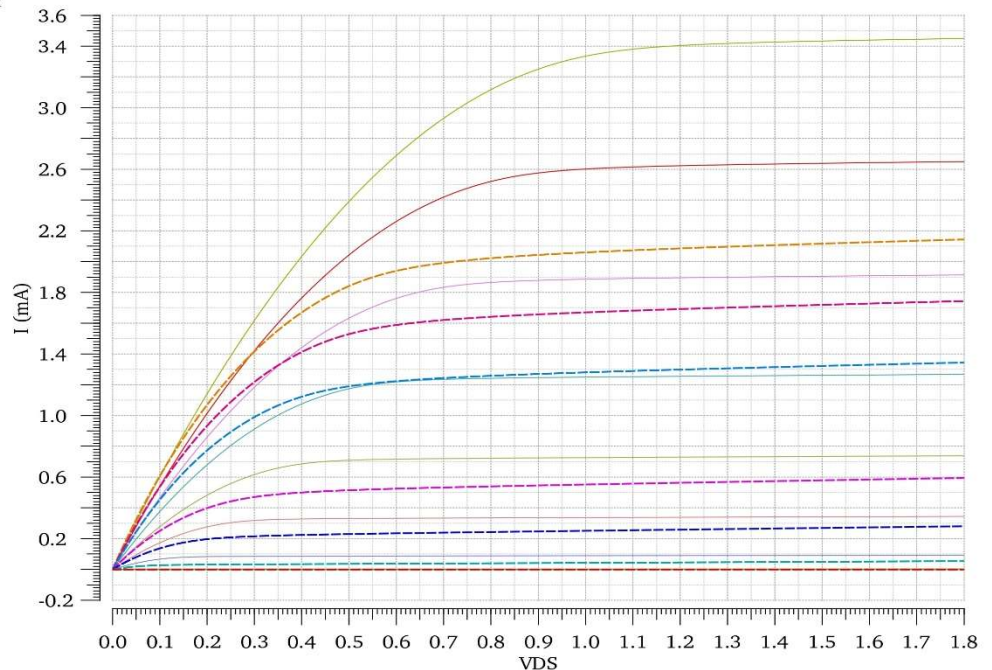
ID vs VDS

DC Response

Sun Jun 21 15:19:09 2026

Name	Vis	LVGS
/M0/D		
/M0/D		2e-07 0
/M0/D		2e-07 0.2
/M0/D		2e-07 0.4
/M0/D		2e-07 0.6
/M0/D		2e-07 0.8
/M0/D		2e-07 1
/M0/D	*	2e-07 1.2
/M0/D		2e-07 1.4
/M0/D		2e-07 1.6
/M0/D		2e-07 1.8
/M0/D		2e-06 0
/M0/D		2e-06 0.2
/M0/D		2e-06 0.4
/M0/D		2e-06 0.6
/M0/D		2e-06 0.8
/M0/D		2e-06 1
/M0/D		2e-06 1.2
/M0/D		2e-06 1.4
/M0/D		2e-06 1.6
/M0/D		2e-06 1.8

Note that Dashed is for short channel



Comment

Current differences

It is clear that the long channel has higher overall current as it doesn't suffer from velocity saturation and can easily reach pinch unlike the short channel. However, sometimes short channel can have higher currents than long channel although suffering from v_{sat} but the smaller rout and other short channel effects like DIBL can help in increasing the current as we can see in the plot in some curves before pinch off the short channel have more current but as soon as v_{sat} hits and it cannot reach pinch off the long channel current becomes bigger.

Slope difference

It can be seen that the short channel slope is larger as the short channel effects like DIBL and stronger CLM helps in increasing the current past saturation and making VGS have less control over the current yielding bigger g_{ds} and more slope.